

NELSPRUIT, South Africa

WMO: 682890

Lat: 25.503S

Lon: 30.912E

Elev: 883

StdP: 91.16

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	7.2	-10.3	1.7	18.4	-7.0	2.3	19.3	12.7	13.9	7.6	14.0	1.6	260	0.292

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.4	32.2	17.7	30.7	18.0	29.3	18.2	22.3	27.2	21.6	26.5	21.0	25.7	3.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	17.3	24.3	20.1	16.5	23.7	19.6	16.0	23.3	70.2	27.7	67.5	26.6	65.4	25.8	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.0	4.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	1.5	23.8	1.7	2.3	0.2	25.4	-0.8	26.8	-1.7	28.1	-3.0	29.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	22.1	22.4	21.9	19.5	17.6	15.5	14.9	16.7	18.9	19.5	20.6	21.3
	DBStd	3.62	2.16	1.85	2.23	2.41	2.58	2.48	2.27	2.87	3.49	3.24	3.02	2.53
	HDD10.0	1	0	0	0	0	0	1	0	0	0	0	0	0
	HDD18.3	410	2	0	3	16	44	90	111	67	36	23	13	6
	CDD10.0	3366	374	348	369	284	236	165	151	209	267	296	317	349
	CDD18.3	733	118	115	114	50	22	4	3	17	53	61	80	96
	CDH23.3	4768	592	580	607	267	171	64	50	228	580	558	562	509
(14) Wind	WSAvg	1.8	1.6	1.8	1.7	1.5	1.7	1.8	1.9	2.1	2.0	2.0	2.0	1.7
	PrecAvg	909	151	135	119	52	17	8	8	13	25	85	137	161
	PrecMax	1383	313	401	192	88	43	40	28	37	91	173	239	322
	PrecMin	393	30	26	14	1	1	1	1	2	3	29	70	71
	PrecStd	218	76	97	50	24	13	10	7	11	22	30	46	54
	DB	31.9	31.6	31.8	30.4	29.0	27.9	27.1	30.4	33.6	34.6	33.5	31.8	
	MCWB	22.0	20.0	19.2	17.4	15.0	12.2	11.9	13.7	15.4	16.7	17.5	20.7	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.6	29.3	29.7	28.0	27.0	25.4	24.8	28.2	31.2	31.3	31.1	29.3	
	2% MCWB	20.8	20.3	19.6	17.7	14.6	12.1	11.7	13.5	14.9	17.0	18.3	20.4	
	5% DB	28.0	28.1	28.1	26.2	25.2	23.2	22.9	26.0	29.2	28.8	28.6	27.7	
	5% MCWB	20.4	19.9	19.6	18.0	14.4	11.8	11.6	13.2	14.9	17.2	19.2	20.1	
	10% DB	26.6	26.8	26.7	24.7	23.5	21.6	21.0	23.9	26.9	26.5	26.4	26.2	
	10% MCWB	19.9	19.7	19.3	17.8	14.4	11.7	11.5	12.8	14.7	16.9	18.8	19.7	
	WB	23.3	22.6	22.2	21.0	18.2	15.1	14.7	16.7	18.2	21.2	25.3	22.7	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.3	28.2	27.8	25.7	24.0	21.8	21.1	24.7	26.6	28.1	25.3	27.9	
	2% WB	22.1	21.7	21.3	20.0	17.1	14.2	13.8	15.8	17.3	19.2	21.4	21.7	
	2% MCDB	27.3	26.8	26.5	24.2	22.5	20.5	19.8	23.2	25.0	26.4	25.7	26.5	
	5% WB	21.4	21.1	20.7	19.2	16.2	13.5	13.1	14.9	16.6	18.4	20.3	21.0	
	5% MCDB	25.9	25.8	25.5	23.3	21.6	19.5	18.9	21.7	24.1	25.4	25.2	25.9	
	10% WB	20.7	20.5	20.1	18.5	15.5	12.9	12.5	14.1	15.8	17.7	19.5	20.3	
	10% MCDB	24.7	24.8	24.6	22.8	20.7	18.7	18.3	20.5	23.2	23.9	24.2	24.8	
(35) Mean Daily Temperature Range	MDBR	7.9	8.4	8.8	9.2	11.5	12.1	11.8	12.1	12.0	10.1	8.9	8.0	
	5% DB	10.4	10.4	11.4	12.3	13.9	14.7	15.2	16.5	16.7	15.5	13.5	10.9	
	MCWBR	4.1	3.7	4.0	4.9	5.3	5.8	5.7	5.8	5.5	5.1	5.1	4.3	
	5% WB	8.7	8.9	9.3	9.9	10.9	11.4	11.2	12.2	12.4	11.9	9.9	9.4	
	MCWBR	4.8	4.0	3.9	4.5	4.7	5.4	5.3	5.2	4.9	4.8	5.2	4.7	
(40) Clear-Sky Solar Irradiance	taub	0.413	0.403	0.398	0.377	0.338	0.321	0.323	0.387	0.441	0.468	0.430	0.431	
	taud	2.333	2.360	2.363	2.382	2.449	2.472	2.455	2.270	2.149	2.126	2.264	2.274	
	Ebn at Noon	932	925	900	873	868	864	876	848	845	856	910	917	
	Edh at Noon	136	130	123	111	94	88	92	122	150	162	145	145	
(44) All-Sky Solar Radiation	RadAvg	6.03	5.96	5.29	4.52	4.20	3.81	4.06	4.70	5.32	5.38	5.72	5.97	
	RadStd	0.47	0.54	0.44	0.33	0.20	0.23	0.14	0.27	0.33	0.38	0.49	0.43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page